Online Retail Sales Analysis

Portfolio Explanation

Project Overview

This project analyses 13 months of transaction data from a UK-based online gift and homeware wholesaler to uncover insights into revenue performance, customer behaviour, product contribution and purchasing patterns over time.

The objective was to demonstrate an end-to-end analytics workflow, from data validation in Excel and cleaning and modelling in MySQL to interactive dashboards in Tableau Public, while applying sound analytical reasoning and verifying every figure before it reached a dashboard or report.

Data Preparation and Modelling

Excel was used for initial validation and rule-based checks before database ingestion. Helper columns flagged missing identifiers, invalid prices, returns and special stock codes without deleting records, and date and time components were separated to support time-based analysis.

The dataset was imported into MySQL using LOAD DATA LOCAL INFILE. During ingestion I resolved regional date parsing (DD/MM/YYYY), character encoding conflicts and MySQL strict-mode constraints by rebuilding the staging layer, enforcing explicit parsing logic and standardising date and time fields from a single source of truth.

A clean analytical table (online_retail_clean) was then built with these rules:

- **Net revenue:** revenue is calculated as quantity multiplied by unit price, so cancelled and returned items are kept as negative lines and deducted rather than removed.
- **Return flag:** an Is_Return field marks cancellation invoices, so order counts and order-based metrics include genuine orders only.
- **Identified customers:** transactions without a customer ID are excluded. Missing IDs had loaded as 0 rather than null, so the rule filters both.
- **Genuine products:** postage, fees and manual adjustment codes are excluded so that product rankings reflect real products.
- **Valid prices:** lines with a zero or negative unit price are excluded.

Validation queries confirmed each rule, and revenue totals were reconciled across every query output (£8,286,663.32 in all cases) before the data was exported to Tableau.

Business Questions Addressed

- How does revenue trend over time?
- Who are the highest-value customers?
- Which products generate the most revenue?
- Which countries contribute the most revenue?
- What is the average revenue per order?
- When do customers place orders during the day?
- How concentrated is revenue across customers (Pareto analysis)?

Each question has its own SQL query, using aggregation, conditional distinct counts and window functions. The query outputs act as an answer key against which every Tableau figure was checked.

Dashboard Design and Insights

Dashboard 1: Executive Overview

Purpose: a concise view of overall performance, seasonality and geographic contribution.

- KPI cards: net revenue (£8.29M), orders (18,402), average order value (£450) and customers (4,334)
- Monthly revenue trend with the September to November peak highlighted and the partial final month annotated
- Top 10 countries with revenue share, usable as a filter for the whole dashboard

Dashboard 2: Customer Insights

Purpose: understand how customer value is distributed and where retention opportunities lie.

- Top 10 customers by net revenue with share of total
- Order frequency bands, showing that 34.7% of customers bought only once
- Pareto curve built with table calculations, annotated to show that 27% of customers generate 80% of revenue

Dashboard 3: Product and Time Insights

Purpose: analyse what customers buy and when they buy it.

- Top 10 products by net revenue with share of total
- Orders and revenue by hour, with the 10:00 to 15:59 window (78.7% of orders) highlighted
- Insights relevant to staffing, campaign timing and operational planning

Tools and Techniques Used

- **Excel:** validation checks and helper columns
- **MySQL:** data loading, cleaning, validation queries, COUNT(DISTINCT CASE ...), window functions with ROWS BETWEEN for running totals
- **Tableau Public:** calculated fields, FIXED level of detail expressions, table calculations (RUNNING_SUM, WINDOW_SUM, INDEX, SIZE), context and Top N filters, dashboard filter actions and annotations
- **Analytical techniques:** time-based analysis, customer segmentation, frequency banding and Pareto (80/20) analysis

Key Takeaways

This project demonstrates the ability to:

- Translate raw transactional data into a reliable analytical model
- Identify and resolve real-world data quality issues, including returns, missing identifiers and non-product records
- Ask meaningful business questions and answer them with data
- Reconcile results across SQL, Tableau and written outputs before publishing
- Build clear, interactive dashboards and turn findings into commercial recommendations

Findings and recommendations are set out in the accompanying Online Retail Business Insights Report.